Journal of Cosmetic and Laser Therapy

ISSN: 1476-4172 (Print) 1476-4180 (Online) Journal homepage: https://www.tandfonline.com/loi/ijcl20

A systematic review of comparative studies of CO_2 and erbium:YAG lasers in resurfacing facial rhytides (wrinkles)

Kee-Hsin Chen, Ka-Wai Tam, I-fan Chen, Shihping Kevin Huang, Pei-Chuan Tzeng, Hsian-Jenn Wang & Chiehfeng (Cliff) Chen

To cite this article: Kee-Hsin Chen, Ka-Wai Tam, I-fan Chen, Shihping Kevin Huang, Pei-Chuan Tzeng, Hsian-Jenn Wang & Chiehfeng (Cliff) Chen (2017) A systematic review of comparative studies of CO_2 and erbium:YAG lasers in resurfacing facial rhytides (wrinkles), Journal of Cosmetic and Laser Therapy, 19:4, 199-204, DOI: 10.1080/14764172.2017.1288261

To link to this article: https://doi.org/10.1080/14764172.2017.1288261

Published online: 07 Apr 2017.

Submit your article to this journal

Article views: 440

View related articles

View Crossmark data

Citing articles: 9 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ijcl20

JOURNAL OF COSMETIC AND LASER THERAPY
2017, VOL. 19, NO. 4, 199–204
http://dx.doi.org/10.1080/14764172.2017.1288261

Taylor & Francis
Taylor & Francis Group

REPORT

A systematic review of comparative studies of CO_2 and erbium:YAG lasers in resurfacing facial rhytides (wrinkles)

Kee-Hsin Chen[a,b,c,d,e], Ka-Wai Tam[d,f,g], I-fan Chen[h], Shihping Kevin Huang[i], Pei-Chuan Tzeng[j], Hsian-Jenn Wang[k,l], and Chiehfeng (Cliff) Chen[d,k,l,m]

[a]Department of Nursing, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan; [b]School of Nursing, College of Nursing, Taipei Medical University, Taipei, Taiwan; [c]Graduate Institute of Clinical Medical Sciences, College of Medicine, Chang Gung University, Taoyuan City, Taiwan; [d]Cochrane Taiwan, Taipei Medical University, Taipei, Taiwan; [e]Evidence-Based Knowledge Translation Center, Wan Fang Medical Center, Taipei Medical University, Taipei, Taiwan; [f]Department of Surgery, Taipei Medical University – Shuang Ho Hospital, New Taipei City, Taiwan; [g]Department of Surgery, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan; [h]Graduate Institute of Management of Technology, National Chiao Tung University, Hsinchu, Taiwan; [i]Institute of Management of Technology, National Chiao Tung University, Hsinchu, Taiwan; [j]Department of Medical Education and Research, Wan Fang Medical Center, Taipei Medical University, Taipei, Taiwan; [k]Division of Plastic Surgery, Department of Surgery, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan; [l]Department of Surgery, School of Medicine, College of Medicine, Taipei Medical University, Taiwan; [m]Department of Public Health, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan

ABSTRACT

Background: Laser resurfacing is used to minimize wrinkles, solar scars and sequelae of acne. **Objective**: Purpose of the systematic review was to compare resurfacing outcomes of CO_2 laser and erbium: yttrium aluminium garnet (erb:YAG) laser therapies. **Materials and methods**: Medline, Cochrane Library, EMBASE and Google Scholar databases were searched until 9 April 2015 using the following terms: laser, carbon dioxide/CO_2, facial wrinkles, rhytides and erbium-doped yttrium aluminium garnet/erbium: YAG/Er:YAG. Two-armed controlled split faced studies that compared CO_2 laser and erbium:YAG laser in patients with mild-to-moderate facial wrinkles or rhytides were included. **Results**: The pooled data in this study and findings of other studies support the greater efficacy with the CO_2 laser in improving facial wrinkles, but the erb:YAG laser was associated with a better complication profile compared with the CO_2 laser. Except one case of hypopigmentation, other complications (i.e., erythema, hyperpigmentation and crusting) and their rates were reported by studies examining both lasers. **Conclusion**: In general, the CO_2 laser appeared to be more efficacious then the erb:YAG laser in treating facial wrinkles. Both lasers treatments were well tolerated.

ARTICLE HISTORY
Received 9 August 2016
Accepted 25 January 2017

KEYWORDS
CO_2; erbium:YAG laser; rhytides; wrinkles

Introduction

Laser resurfacing is used to minimize wrinkles, solar scars and sequelae of acne (1,2). High-energy pulsed (10,600 nm) scanning carbon dioxide (CO_2) lasers are employed to remove rhytides and other effects of photo damage and acne damage (2–4). The CO_2 laser uses coagulative thermal damage for skin resurfacing and rejuvenation (5). The laser-tissue interaction is typically short (<1 ms) resulting in limited thermal diffusion during a pulse (5). However, a drawback of the CO_2 laser is that the treatment is associated with adverse effects, particularly dyschromia and prolonged erythema associated with photothermolysis (6–8). The level of erythema and time for re-epithelialization can keep patients at home for more than 2 weeks (3).

The erbium:yttrium aluminium garnet (erb:YAG) laser that uses lower energy (2940 nm) is also used in facial resurfacing. It has the advantage that it ablates tissue with little thermal damage (9,10). The residual thermal damage with the erb: YAG laser is about 50 µm compared with 80–150 µm commonly seen with multiple passes of pulsed CO_2 laser exposure (3). The 2940 wavelength is closer to the absorption peak of water (3000 nm) than that of the CO_2 laser, resulting in most of the energy being absorbed in the epidermis and papillary dermis; consequently, there is more superficial ablation with less thermal damage (11). The erb:YAG laser has been found to associate with shorter post-operative recovery period and less erythema than the CO_2 laser (12).

There have only been a very limited number of systematic reviews or meta-analyses that compared the efficacy and side effects of pulsed CO_2 and erb:YAG lasers. The purpose of this systematic review was to present and compare outcomes of the two laser therapies for resurfacing and collagen remodelling in treating facial skin damage.

Methods

Search strategy

This systematic review was performed according to the PRISMA guidelines. Medline, Cochrane Library, EMBASE and Google Scholar databases were searched until 9 April 2015 using the following search terms: laser, carbon dioxide/CO_2, facial

CONTACT Chiehfeng (Cliff) Chen clifchen@tmu.edu.tw Department of Public Health, School of Medicine, College of Medicine, Taipei Medical University, No. 111, Section 3, Hsing-Long Rd, Taipei 116, Taiwan.

© 2017 Taylor & Francis

wrinkles, rhytides, erbium-doped yttrium aluminium garnet/erbium:YAG/Er:YAG. Two-armed controlled split faced studies that compared CO_2 and erb:YAG lasers in patients with mild-to-moderate facial wrinkles or rhytides were included. One-arm studies were excluded, as were letters, comments, editorials, case reports, proceeding and personal communication. Only English publications were included. Study eligibility was determined by two independent reviewers. When there was uncertainty regarding eligibility, a third reviewer was consulted.

Data extraction

The following information/data were extracted from studies that met the inclusion criteria: the name of the first author, year of publication, study design, number of participants in each study group, participants' age and gender, skin conditions and facial wrinkle types, types of intervention and laser parameters used for each study/control group and the outcomes (efficacies reported and rates of various complications).

Results

Search results

The initial search identified 42 potential studies of which 36 were excluded for either not being relevant or being duplicates, the study used a different type of laser than that of interest, the study only evaluated one type of laser or the report was a commentary (Figure 1).

Six studies were included in the systematic review. All six studies were split-face randomized controlled trials (RCT, see Table 1) (3,13–17). Across the studies, the population of patients was small, ranging from 13 to 28 patients. The mean age ranged from 19 to 90 years, and most patients were women. The location of the laser treatment was primarily in the periorbital or perioral region in five of the studies (3,14–17), with one study treating the upper lip region (13). The types of wrinkles were mild to moderate, and ranged from Fitzpatrick type I to type IV. The energy of lasers used and spot areas are described in Table 1. The follow-up time across the studies was 2–6 months.

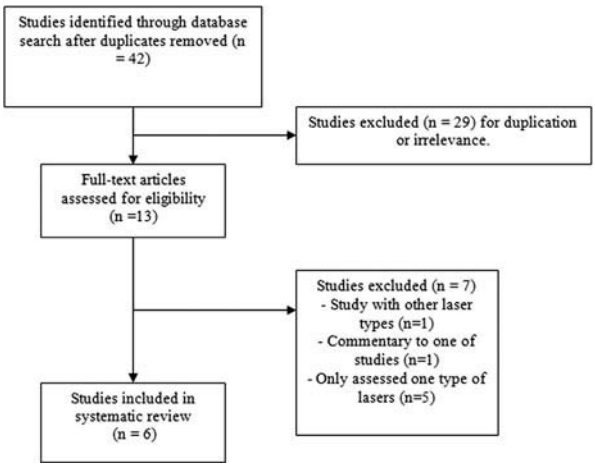

Figure 1. Flow chart for the study selection.

Several outcomes of laser treatment were evaluated including post-operative erythema, hypopigmentation, hyperpigmentation, crusting, Fitzpatrick wrinkle score following treatment and degree of improvement (Table 2). Three of the six studies reported findings on post-operative erythema (3,14,17). Ross et al. found that the CO_2 laser therapy was associated with lower-grade erythema (grade 2.7 at two weeks and 1.5 at six weeks) compared with erb:YAG laser treatment (grade 3.4 at two weeks and 1.8 at six weeks). However, Khatri et al. found that a greater percentage of patients with CO_2 laser treatment experienced erythema at two and six weeks (95% and 62%, respectively) compared with erb:YAG laser therapy (67% and 24%). McDaniel et al. found that with both lasers, the erythema was reversible, and the duration was comparable between therapy groups (9.8 weeks each). These findings indicate that the CO_2 laser therapy had a higher frequency of erythema, although of lower grade, compared with erb:YAG.

Only one study reported hypopigmentation, and the incidence was higher with CO_2 laser treatment (43%) then erb:YAG laser therapy (5%) (Table 2) (3). Three of the six studies reported hyperpigmentation, and the incidence and severity were similar between treatments (3,14,16). In the three studies that reported findings on crusting, the intensity and length of crusting was greater with CO_2 than erb:YAG laser therapy (3,13,16). Both the hyperpigmentation and crusting were reversible for both laser treatments. Overall, CO_2 laser therapy was associated with a higher degree of wrinkle improvement than erb:YAG laser treatment across all the studies (Tables 2, 3 and Figure 2).

Discussion

In this study, the erb:YAG laser was associated with a slightly better complication profile compared with the CO_2 laser. The CO_2 laser was associated with a higher incidence of erythema, although of lower grade, compared with the erb:YAG laser, and with a greater frequency and intensity of crusting. The lasers were similar in the incidence of hyperpigmentation. Except for one case of hypopigmentation, other complications (i.e., erythema, hyperpigmentation and crusting) were reversible following treatment with either laser. Hyperpigmentation is a common longer-term adverse effect of laser resurfacing treatment with both lasers (18). Crusting and erythema are unavoidable short-term side effects of fractional laser treatments (18). It is not possible to statistically compare the findings among the studies due to a lack of homogeneity in the methods of reporting results used by included studies.

In the three studies that provided comparable data between the two laser treatments, the CO_2 laser was associated with greater efficacy than the erb:YAG (3,12,13). The trend for better efficacy with the CO_2 laser treatment was also apparent in the other three studies, but authors of those three studies concluded that the two treatments were similar in their ability to treat facial wrinkles (14,15,17). Of note, the study of McDaniel et al. compared CO_2 laser treatment to the combined therapy of CO_2 and erb:YAG lasers, which complicates the comparison of the study's findings with those of the other four studies. Various types of treatments have been used for facial

Table 1. Summary of basic characteristics for selected studies.

First author (year)	Study design	Number of patients	Age (years)	Women (%)	Facial wrinkles (location and types): Locations	Facial wrinkles (location and types): Types/depth	Intervention: Type of laser	Intervention: Energy of laser	Spot diameter (mm)	Length of follow-up (months)
Karsai (2010)	Double-blind split-face RCT	28	Mean = 46.1 (SD = 4.0)	92.9	Periorbital	Fitzpatrick type I–II	Ablative factional CO_2	2 pass (15 mJ × 1 pass + 20 mJ × 1 pass)		3
							Ablative fractional Er:YAG	4 passes		
Ross (2001)	Prospective split-face RCT	13	Mean = 47 (range: 30–80)	92.3	Periorbital (n = 5), perioral (n = 5), and both (n = 3)	Fitzpatrick type I–III	Ablative CO_2	1 pass	1	6
							Ablative Er:YAG	4 passes	3–4	
Newman (2000)	Split-face RCT	21	Mean = 53 (range: 39–74)	100	Upper lip region	Fitzpatrick type I–IV	Ablative CO_2	300 mJ × 2 passes	2.25	2
							Ablative Er:YAG	6 passes (1.5 J × 4 passes 2.6 J × 1 pass 1.5 J × 1 pass)		
Adrian (1999)	Split-face RCT	20	Range: 39–72	n/a	Perioral (n = 12)	Fitzpatrick II–III	Ablative CO_2	300 mJ × 3 passes		3
							Ablative Er:YAG	12–15 passes	5	
					Periorbital (n = 8)		Ablative CO_2	300 mJ × 3 passes (2 passes with 300 mJ of energy to the lower preseptal tissue and 2 or 3 passes to the 'crow's feet' area)	n/a	
							Ablative Er:YAG	(5–7 pass were delivered to the lower eyelid skin, 7–10 passes were delivered to the lateral 'crow's feet' area)	5	
Khatri (1999)	Split-face RCT	21	Range: 18–90	90.5	Perioral (n = 12), periorbital (n = 9)	Fitzpatrick I–III	Ablative CO_2	2–3 passes (450 mJ 1st pass, then 350 mJ in subsequent passes)	n/a	6
							Ablative Er:YAG	<5 passes (1–1.5 J/pass)	n/a	
							Ablative Er:YAG	5 passes or more	n/a	
McDaniel (1999)	Split-face RCT	20	Mean = 59 (range: 42-72)	100	Perioral	Fitzpatrick type I–III	Ablative CO_2	300 mJ × 2 pass	2.25	4
							Ablative CO_2 + Er:YAG	300 mJ (×2 passes) + 2.0 J (×3 passes)	2.25	

Abbreviations: RCT, randomized controlled trials; SD, standard deviation; n/a, not available.

Table 2. Summary of interventions and outcomes for selected studies.

First author (year)	Number of patients	Laser types	Post-operative erythema	Hypopigmentation	Hyperpigmentation (0–3 months)	Duration of crusting (days)	Fitzpatrick wrinkle score baseline => last visit	Degree of improvement
Karsai (2010)	28	CO_2		0	n/a	n/a	4.76 (1.3) => 4.27 (1.1)	9.7%
		Er:YAG		0	n/a	n/a	4.63 (1.4) => 4.31 (1.3)	7.4%
Ross (2001)	13	CO_2	Grade of erythema: 2.7 (2 wks), 1.5 (6 wks)	0	4/8 in periorbital group, 3/8 in perioral group (severity score 1.1 for CO_2, 1.3 for Er:YAG) at 3 months	n/a	4.2 (2.1) => 3.1 (1.7)	26.2%
		Er:YAG	Grade of erythema: 3.4 (2 wks), 1.8 (6 wks)	0	n/a	n/a	4.2 (2.2) => 3.2 (1.5)	23.8%
Newman (2000)	21	CO_2	n/a	0	n/a	7.8	n/a	63%
		Er:YAG	n/a	0	1 (Asian subset)	3.5	n/a	54%
Adrian (1999)	20	CO_2	n/a	n/a	n/a	10–12	n/a	75–100%: 12/12
		Er:YAG	n/a	n/a	n/a	8–10	n/a	25–50%: 2/12; 50–75%: 5/12; 75–100%: 5/12
		CO_2	n/a	n/a	n/a	n/a	n/a	50–75%: 3/8; 75–100%: 5/8
		Er:YAG	n/a	n/a	n/a	n/a	n/a	25–50%: 2/8; 50–75%: 4/8; 75–100%: 2/8
Khatri (1999)	21	CO_2	Erythema: at 2 wks: 20 (95%), at 2 months: 13 (62%)	9 (43%)	6 (29%)	n/a	n/a	Score 4 out of 5
		Er:YAG	Erythema: at 2 wks: 14 (67%), at 2 months: 5 (24%)	1 (5%)	5 (24%)	n/a	n/a	Score 2 of 5
		Er:YAG				n/a	n/a	Score 4 of 5
McDaniel (1999)	20	CO_2	Duration of erythema = 9.8 wks	0	n/a	7.4	n/a	67.3
		CO_2 + Er:YAG	Duration of erythema = 9.8 wks	0	n/a	6.5	n/a	68.5

Wrinkle score was represented as mean (SD).
SD, standard deviation; wks, weeks; n/a, not available.

Table 3. Reported degrees of improvement from different treatments at given locations.

First author (year)	Number of patients	Location	Degree of improvement	
			CO_2	Er:YAG
Ross (2001)	13	Periorbital (n = 5), perioral (n = 5), and both (n = 3)	26.20%	23.80%
Adrian (1999)	12	Perioral	75–100%: 12/12	25–50%: 2/12; 50–75%: 5/12; 75–100%: 5/12
	8	Periorbital	50–75%: 3/8; 75–100%: 5/8	25–50%: 2/8; 50–75%: 4/8; 75–100%: 2/8
Khatri (1999)	21	Perioral (n = 12), periorbital (n = 9)	Score 4 out of 5	Score 2 of 5 (<5 passes), score 4 of 5 (≥5 passes)

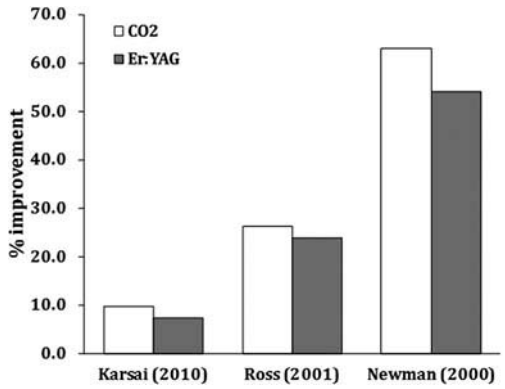

Figure 2. Improvement assessed and reported by included studies for patients treated with CO_2 and Er:YAG lasers. Data were presented as percentages of improvement based on visual assessment before and after laser treatments.

rejuvenation of damaged skin including topical retinoids, bleaching agents, chemical peeling and dermabrasion (18). Laser therapies, such as high-energy pulsed or scanned CO_2 lasers and erb:YAG lasers, are equal to or better than prior treatments (18). With CO_2 and erb:YAG lasers, the incidence of long-term complications has been low, particularly in experienced hands (19,20). For CO_2 lasers, the short-term complications include transient erythema, pruritus, milia formation, acneiform, pustules and transient scarring. Hypopigmentation is sometimes observed in CO_2 laser treatment (18), and is a long-term complication that can potentially become irreversible. A major complication associated with erb:YAG is the transient hyperpigmentation, which is typically short term in nature (18). Hyperpigmentation appears to be related to the ablation depth, which is less with the erb:YAG than the CO_2 laser. In general and as seen in our study, CO_2 lasers are associated with higher incidences of post-surgery morbidity than erb:YAG (18). There are also combined erb:YAG/CO_2 dual-mode laser systems available, notably the Derma K (Lumenis, Santa Clara, California) and the Sciton Contour (Palo Alto, California), which supposedly combine the advantages of both laser systems, but controlled comparative studies of these systems are still lacking.

This systematic review has several strengths. All included studies were split-face RCTs that evaluated same patients with similarly mild-to-moderate facial wrinkles (ranged Fitzpatrick scores I–IV, but most were I–III). Among included studies, patients with lesions of malignancy or hypertrophic keloid, or scar formation were typically excluded. However, there are some limitations to this study that should be considered. The number of included studies was small. In addition, the studies differed in the method used to evaluate efficacy, with a mixture of subjective and objective criteria. This heterogeneity precluded the ability to perform a meta-analysis. The method of reporting adverse effects also differed across the studies, making a statistical comparison between treatments impossible. There was also heterogeneity in the laser parameters (laser energy, spotting size, fluencies, number of passes, etc.) used among the studies, and the skill of physicians who performed the laser treatment may have varied across the studies, which could have impacted both efficacy and complication rates. Lastly, Karsai et al. (15) describe the type of laser used (ablative fractional) clearly, whereas the other studies did not specifically state the type of laser. However, we believe we can assume that the other studies used ablative nonfractional lasers (21,22).

In conclusion, the CO_2 laser appears to have better efficacy than the erb:YAG laser in treating facial wrinkles. Both lasers were well tolerated with the erb:YAG being associated with fewer reported incidences of complication.

Funding

This study was supported by Taipei Medical University – Wan Fang Hospital under the Grant Number 106-eva-32.

Conflict of interest

The authors declare that there is no conflict of interest.

References

1. de Noronha L, Chin EW, Menini CM, Knopfholz J, Rampazzo JC, Graf R. Histopathologic and morphometric evaluation of the skin abnormalities induced by erbium: YAGand carbon dioxide lasers in 10 patients. Plast Reconstr Surg. 2001;108(5):1380–1388.
2. Fitzpatrick RE, Goldman MP, Satur NM, Tope WD. Pulsed carbon dioxide laser resurfacing of photo-aged facial skin. Arch Dermatol. 1996;132(4):395–402.
3. Khatri KA, Ross V, Grevelink JM, Magro CM, Anderson RR. Comparison of erbium: YAGand carbon dioxide lasers in resurfacing of facial rhytides. Arch Dermatol. 1999;135(4):391–397.
4. Ransom ER, Antunes MB, Bloom JD, Greco T. Concurrent structural fat grafting and carbon dioxide laser resurfacing for perioral and lower face rejuvenation. J Cosmet Laser Ther. 2011;13(1):6–12.
5. Kauvar AN, Waldorf HA, Geronemus RG. A histopathological comparison of "char-free" carbon dioxide lasers. Dermatol Surg. 1996;22(4):343–348.
6. Goldman MP, Manuskiatti W. Combined laser resurfacing with the 950-microsec pulsed CO2 + Er: YAGlasers. Dermatol Surg. 1999;25(3):160–163.

7. Goodman GJ. The limitations of skin resurfacing techniques. The necessity to combine procedures. Dermatol Surg. 1998;24(6):687–688.
8. Nanni CA, Alster TS. Complications of cutaneous laser surgery. A review. Dermatol Surg. 1998;24(2):209–219.
9. Kaufmann R, Hartmann A, Hibst R. Cutting and skin-ablative properties of pulsed mid-infrared laser surgery. J Dermatol Surg Oncol. 1994;20(2):112–118.
10. Kaufmann R, Hibst R. Pulsed Erbium:YAG laser ablation in cutaneous surgery. Lasers Surg Med. 1996;19(3):324–330.
11. Riggs K, Keller M, Humphreys TR. Ablative laser resurfacing: high-energy pulsed carbon dioxide and erbium:yttrium-aluminum-garnet. Clin Dermatol. 2007;25(5):462–473.
12. Adrian RM. Pulsed carbon dioxide and erbium-YAG laser resurfacing: a comparative clinical and histologic study. J Cutan Laser Ther. 1999;1(1):29–35.
13. Newman JB, Lord JL, Ash K, McDaniel DH. Variable pulse erbium: YAGlaser skin resurfacing of perioral rhytides and side-by-side comparison with carbon dioxide laser. Lasers Surg Med. 2000;26(2):208–214.
14. Ross EV, Miller C, Meehan K, Pac A, McKinlay J, Sajben P, et al. One-pass CO2 versus multiple-pass Er: YAGlaser resurfacing in the treatment of rhytides: a comparison side-by-side study of pulsed CO2 and Er: YAGlasers. Dermatol Surg. 2001;27(8):709–715.
15. Karsai S, Czarnecka A, Junger M, Raulin C. Ablative fractional lasers (CO(2) and Er: YAG):a randomized controlled double-blind split-face trial of the treatment of peri-orbital rhytides. Lasers Surg Med. 2010;42(2):160–167.
16. Adrian RM. Pulsed carbon dioxide and long pulse 10-ms erbium-YAG laser resurfacing: a comparative clinical and histologic study. J Cutan Laser Ther. 1999;1(4):197–202.
17. McDaniel DH, Lord J, Ash K, Newman J. Combined CO2/erbium: YAGlaser resurfacing of peri-oral rhytides and side-by-side comparison with carbon dioxide laser alone. Dermatol Surg. 1999;25(4):285–293.
18. Papadavid E, Katsambas A. Lasers for facial rejuvenation: a review. Int J Dermatol. 2003;42(6):480–487.
19. Nanni CA, Alster TS. Complications of carbon dioxide laser resurfacing. An evaluation of 500 patients. Dermatol Surg. 1998;24(3):315–320.
20. Bernstein LJ, Kauvar AN, Grossman MC, Geronemus RG. The short- and long-term side effects of carbon dioxide laser resurfacing. Dermatol Surg. 1997;23(7):519–525.
21. Manstein D, Herron GS, Sink RK, Tanner H, Anderson RR. Fractional photothermolysis: a new concept for cutaneous remodeling using microscopic patterns of thermal injury. Lasers Surg Med. 2004;34(5):426–438.
22. Preissig J, Hamilton K, Markus R. Current laser resurfacing technologies: a review that delves beneath the surface. Semin Plast Surg. 2012;26(3):109–116.